TRAINET

Undertaken By:

MAHEEN RAZA
REG. NO. CIIT/FA22-BSE-086/WAH

TEHREEM MIRZA
REG. NO. CIIT/FA22-BSE-088/WAH

Supervised By**:**
Mr. MUHAMMAD IBRAHIM

DEPARTMENT OF COMPUTER SCIENCES

COMSATS UNIVERISTY ISLAMABAD, WAH CAMPUS

PAKISTAN

SESSION 2022-2026

Table of Contents

1 INTRODUCTION 1

1.1 SYSTEM INTRODUCTION 1

1.2 BACKGROUND OF THE SYSTEM 1

1.3 OBJECTIVES OF THE SYSTEM 2

1.4 SIGNIFICANCE OF THE SYSTEM 2

2 OVERALL DESCRIPTION 4

2.1 PRODUCT PERSPECTIVE 4

2.2 PRODUCT SCOPE 5

2.3 PRODUCT FUNCTIONALITY 5

2.4 USERS AND CHARACTERISTICS 7

2.5 OPERATING ENVIRONMENT 8

3 SPECIFIC REQUIREMENTS 10

3.1 FUNCTIONAL REQUIREMENTS 10

3.2 BEHAVIOUR REQUIREMENTS 15

3.3 EXTERNAL INTERFACE REQUIREMENTS 18

3.3.1 User Interfaces 18

3.3.2 Hardware Interfaces 20

3.3.3 Software Interfaces 20

3.3.4 Communications Interfaces 22

4 OTHER NON-FUNCTIONAL REQUIREMENTS 24

4.1 PERFORMANCE REQUIREMENTS 24

4.2 SAFETY AND SECURITY REQUIREMENTS 24

4.3 SOFTWARE QUALITY ATTRIBUTES 25

4.3.1 Reliability 25

4.3.2 Usability 26

4.3.3 Maintainability 26

4.3.4 Scalability 26

4.3.5 Portability 26

4.3.6 Interoperability 26

4.3.7 Security (Quality-Level) 26

4.3.8 Design for Change 27

5 DESIGN DESCRIPTION 28

5.1 COMPOSITE VIEWPOINT 28
5.2 LOGICAL VIEWPOINT 28
5.3 INFORMATION VIEWPOINT 30
5.4 INTERACTION VIEWPOINT 30
5.5 STATE DYNAMICS VIEWPOINT 31
5.6 ALGORITHM VIEWPOINT 32

Table of Figures

FIGURE 1: SYSTEM CONTEXT DIAGRAM OF TRAINET 4
FIGURE 2: USE CASE DIAGRAM OF TRAINET 15
FIGURE 3: PACKAGE DIAGRAM OF TRAINET 28
FIGURE 4: CLASS DIAGRAM OF TRAINET 29
FIGURE 5: ENTITY RELATIONSHIP DIAGRAM OF TRAINET 30
FIGURE 6: SEQUENCE DIAGRAM OF TRAINET 31
FIGURE 7: SEQUENCE DIAGRAM OF TRAINET 32
FIGURE 8: SEQUENCE DIAGRAM OF TRAINET 33
FIGURE 9: SEQUENCE DIAGRAM OF TRAINET 34
FIGURE 10: SEQUENCE DIAGRAM OF TRAINET 34
FIGURE 11: SEQUENCE DIAGRAM OF TRAINET 35
FIGURE 12: STATE MACHINE DIAGRAM OF TRAINET 36
FIGURE 13: STATE MACHINE DIAGRAM OF TRAINET 36
FIGURE 14: STATE MACHINE DIAGRAM OF TRAINET 37
FIGURE 15: STATE MACHINE DIAGRAM OF TRAINET 37

1 Introduction

1.1 System Introduction

TRAINET is a modern web-based learning and career development platform. The goal of it is to bridge the gap between academic education and practical industry demands. It combines industry connection, project-based skill development, secure certificate verification, and AI-powered learning into a single, cohesive system.

With the goal of preparing students for the workforce, the platform offers professional mentoring opportunities, intelligence assessment tools, arranged online courses, and real-world project experiences.

TRAINET offers a comprehensive digital environment for students, trainers, graduates, and recruiters by integrating traditional learning elements with modern technology, including AI auto-grading, plagiarism detection, QR-verified certificates, alumni assistance, and an AI-driven talent pool.

1.2 Background of the System

A lot of the current Learning Management Systems (LMS) are primarily concerned with material delivery and simple evaluations. They do not, however, offer practical experience, secure verification, or recruiter connections. Recruiters struggle to find skill-based candidates, institutions struggle to ensure the authenticity of certificates, and students frequently graduate without practical skills.

Unlike traditional LMS platforms, TRAINET introduces:

- AI-integrated learning (auto-grading, plagiarism detection, feedback)
- QR-based secure certificate verification
- Real-world Work & Practice tasks

- Alumni & Consultancy Network
- AI Talent Pool for skill-based hiring

These characteristics differentiate TRAINET from traditional educational systems and make it a modern, hybrid platform that focus on learning, practical experience, and and preparing learners to stand out to recruiters.

1.3 Objectives of the System

- Provide AI-powered learning through automated grading, plagiarism detection, and intelligent feedback.
- Provide safe registration for courses and QR code-based certificate verification.
- Encourage practical learning through industrial challenges, coding assignments, and mini projects.
- Provides an alumni and consulting network to provide career counselling and mentoring.
- AI-driven skill matching Connect learners with recruiters.
- Facilitate learning methods that are adaptable and based on performance and skill analysis.
- Provide an integrated platform for sharing materials, conducting live sessions, delivering courses, and monitoring progress.
- Provide administrators with dashboards and analytics to keep an eye on system-wide activity.

1.4 Significance of the System

TRAINET is significant because it:

- Connects academic knowledge with industrial demands.
- Guarantee trust and authenticity through secure QR-based verification.
- Increase productivity through integrated talent pools and real-world Tasks.
- Establishes a cooperative learning environment with companies, trainers, and alumni.
- Encourages work based on skills rather than degrees.

- Provides data analytics to organizations so they may make well-informed decisions.
- Uses practice assignments and mini projects to assist learners in developing strong portfolios.
- Universities, training facilities, online academies, professional boot camps, and corporate upskilling programs can all use TRAINET.

2 Overall Description

2.1 Product Perspective

TRAINET is a stand-alone web-based software solution intended to serve as a comprehensive platform for learning, training, practice, and hiring. It is a new, standalone solution that integrates several modules into a single digital ecosystem rather than an expansion of an earlier system.

The system links with a number of external services, including third-party QR code scanning apps for certificate verification and Google Meet for live sessions. All user roles, including recruiters, administrators, alumni, trainers, and students, function internally inside an organized set of linked modules. The system can smoothly provide learning, verification, mentorship, and recruitment functionalities because each module adds to the overall workflow.

SYSTEM CONTEXT DIAGRAM

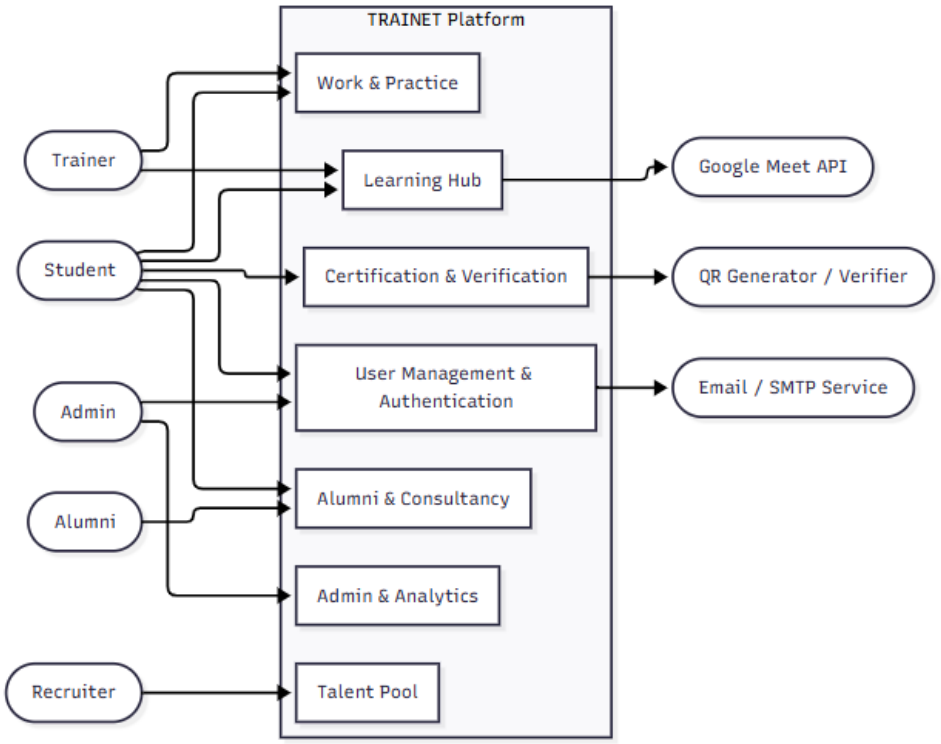

Figure 1: System context diagram of TRAINET

2.2 Product Scope

TRAINET offers a complete digital environment that facilitates education, the development of practical skills, alumni involvement, certificate verification, and talent identification. In addition to providing a safe and dependable method for academic and professional verification, the system focuses on allowing students to obtain practical experience using AI-supported tools and organized project activities.

In Scope:

- Enrolment in courses, exchange of materials, and monitoring of progress
- AI-powered plagiarism detection and auto-grading
- Registration for courses and certificate verification using QR codes
- Google Meet-integrated live sessions with capabilities for alumni mentoring and consulting
- Tasks for work and practice, such as coding challenges and mini-projects
- AI-based skill matching for recruiters' talent pool
- System-wide analytics and administrative dashboards
- Role-based access control and user authentication

Out of Scope:

- Offline training or in-person instruction in a classroom
- Payroll, HR software, or services unrelated to training
- Features of social networking sites that are not educational
- Documentation related to external recruiting (CV screening outside platform)
- Any implementation relying on hardware

The scope makes it abundantly clear that TRAINET is a web-based system for career development and education; it is not an alternative for social networking tools, HRMS, or generic LMS.

2.3 Product Functionality

 User Management & Authentication

- Secure sign-up and login

- Role-based interface and permissions
- QR-based enrollment

 Learning Hub

- Course content delivery
- Live sessions
- AI-enabled assignment evaluation and plagiarism detection

 Certification & Verification

- Automated certificate generation
- QR-embedded verification system

 Alumni & Consultancy Network

- Alumni profiles and achievements
- Mentorship and consultancy sessions
- Professional communication between learners and alumni

 Work & Practice

- Mini-projects and real-world tasks
- Industry-level challenges posted by trainers
- AI auto-grading and peer-review options

 Talent Pool (Recruiter Module)

- AI-based matching between recruiters and learners
- Skill-based search and portfolio viewing

 AI Personalization & Recommendation

- Adaptive learning paths
- Skill-gap analysis
- Personalized course recommendations

 Admin & Analytics

- Administrative controls
- Metrics and reports for decision-making

2.4 Users and Characteristics

User Type	Description & Role	Primary Needs
Student	Primary system learner. Participate in classes, turns in assignments, checks certificates, and completes Work & Practice assignments.	• Simple navigation and enrolment for courses • Access to educational resources and live sessions • AI-powered plagiarism detection and assignment checking; • certificates with QR code verification • Exposure to opportunities and tasks in the real world • Talent Pool access to recruiters
Trainer	Experts in the field are in charge of overseeing course progress, posting challenges, delivering course content, and assessing submissions.	• Uploading and organizing course materials. • publishing coding assignments and mini-projects • having access to AI auto-grading assistance • Monitoring student development and performance • Effective communication strategies for learners
Alumni	Qualified professionals and	• Management of profiles and portfolios

	graduates who provide students with career advice, mentorship, and consultation.	• A willingness to offer guidance or advice • Features for helping learners via communication • Visibility in the Alumni Network and Talent Pool
Recruiter	External experts and organizations use AI-based matching and portfolio evaluation to find qualified applicants.	• Tools for skill-based searching and filtering • Verified student portfolios and project work are accessible. • AI-powered talent matching • A trustworthy way for accessing candidates
Admin	Oversees system operations, controls user responsibilities, produces analytics reports, and guarantees the integrity and security of the system.	• Complete system configuration and control • Dashboards and analytics accessibility. • The capacity to oversee every kind of user • Tools for keeping an eye on platform security and performance • Tools for reporting institutional decisions

Table 1: Users and Characterstics of TRAINET

2.5 Operating Environment

TRAINET will function in an online environment that facilitates mobile and web access. The system is intended to operate via common web browsers on internet-connected devices, including laptops, desktop computers, tablets, and smartphones. Because TRAINET offers real-time features including live sessions, AI-driven assessments, progress synchronization, certificate verification, and project submissions, a stable internet connection is necessary.

All contemporary operating systems, including Windows, macOS, Linux, iOS, and Android, will support the platform with ease. To guarantee high availability, data consistency, and dependability, all backend data, user profiles, course materials, and system functions will be housed on a secure cloud-based server. While maintaining system speed and data integrity, the environment will be able to manage several concurrent users, including administrators, trainers, alumni, recruiters, and students.

Minimum platform requirements include:

- **Web Browsers:** Chrome, Firefox, Edge, or Safari (latest versions).
- **Internet Connection:** Stable broadband or mobile data connection.
- **Supported Devices:** Desktop computers, laptops, tablets, and Android/iOS smartphones.
- **Operating Systems:** Windows, Linux, Android, iOS.
- **Server Environment:** Cloud-hosted web server with database support.

This setting guarantees TRAINET's continued availability, scalability, security, and dependability for all users on diverse platforms.

3 Specific Requirements

3.1 Functional Requirements

The primary system functions and operations that TRAINET must carry out are outlined in this section. To ensure that the platform serves as a comprehensive learning, training, and recruitment ecosystem, each functional requirement supports one or more system objectives.

User Management & Authentication Module Requirements

ID	Description
FR-UM-1	The system will allow all type of user (learners, trainers, alumni, recruiters, and administrators) to register and create accounts through registration form.
FR-UM-2	The system will provide secure login functionality using email/username and password.
FR-UM-3	The system will support role-based access control based on assigned user type.
FR-UM-4	The system will be able to send verification emails or OTP codes for validation of account.
FR-UM-5	The system will allow learners to enroll in courses by using QR-based registration.
FR-UM-6	Users are allowed to update their profile information on the system.
FR-UM-7	The system will limit access to unauthorized modules based on the roles of the user.

Table 2: Functional Requirements of TRAINET

Learning Hub Module Requirements

ID	Description
FR-LH-1	Trainers will be able to add resources, slides, PDFs, videos, and other course materials into the system.
FR-LH-2	The system will allow learners to access their course material and monitor their progress.
FR-LH-3	The system will allow integration of Google meet links for live class sessions.
FR-LH-4	The system will allow trainers to assign tasks, assignments and quizzes.
FR-LH-5	The system will allow AI-powered evaluation of assignments.
FR-LH-6	The system will run plagiarism detection for all the uploaded assignments by learners.
FR-LH-7	For student submissions the system will offer comments/feedbacks (either trainer provided or AI-generated).

Table 3: Functional Requirements of TRAINET

Certification & Verification Module Requirements

ID	Description
FR-CV-1	After the course completion the system will provide learners with a digital certificates
FR-CV-2	A unique QR code will be provided on each certificate for it verification.
FR-CV-3	The system will allow user to scan a QR code on certificate to check its authenticity.
FR-CV-4	Certificate details will be displayed after successful verification.

FR-CV-5	For auditing reasons, the system will record every certificate production and verification event.

Table 4: Functional Requirements of TRAINET

Alumni & Consultancy Network Module Requirements

ID	Description
FR-AL-1	Alumni will be able to create professional profiles on the system that include portfolios, accomplishments, and experience.
FR-AL-2	The learners will be able to view alumni profiles on the system if they want assistance and mentorship.
FR-AL-3	The system will allow alumni to provide mentorship and consulting sessions to the learners.
FR-AL-4	The system will offer learners and alumni a communication channel.
FR-AL-5	Alumni will be able to update or modify their professional information on the system.

Table 5: Functional Requirements of TRAINET

Work & Practice Module Requirements

ID	Description
FR-WP-1	The trainers will be allowed by the system to post work and practice tasks such as mini projects, coding tasks, and challenges.
FR-WP-2	Learners will be allowed by the system to submit their solutions of assigned tasks
FR-WP-3	AI-based auto grading will be done to the solutions submitted by learners.
FR-WP-4	Learners can see their results, scores and trainer comments.

Table 6: Functional Requirements of TRAINET

Talent Pool & Recruiter Module Requirements

ID	Description
FR-TP-1	The system will allow recruiters to search for candidates based on performance, skills, results, and abilities.
FR-TP-2	The system will facilitate AI-based candidate matching on requirements of recruiters.
FR-TP-3	The system will allow recruiters to view learner's certificates and results.
FR-TP-4	The system will provide a communication channel between recruiters and candidates.
FR-TP-5	Recruiters will be able to save or bookmark candidates.

Table 7: Functional Requirements of TRAINET

AI Personalization & Recommendation Module Requirements

ID	Description
FR-AI-1	The system will analyze learner skills and performance.
FR-AI-2	The system will recommend learners courses based on their interests and demand.

Table 8: Functional Requirements of TRAINET

Admin & Analytics Module Requirements

ID	Description
FR-AD-1	The system will provide admin with a dashboard course statistics, number of users, and activity metrics.
FR-AD-2	The system will allow administration to manage course tasks, users, and certifications.

FR-AD-3	The system will keep an eye on system events and security records.
FR-AD-4	The administrator will be able to modify system settings.

3.2 Behaviour Requirements

Use Case Diagram:

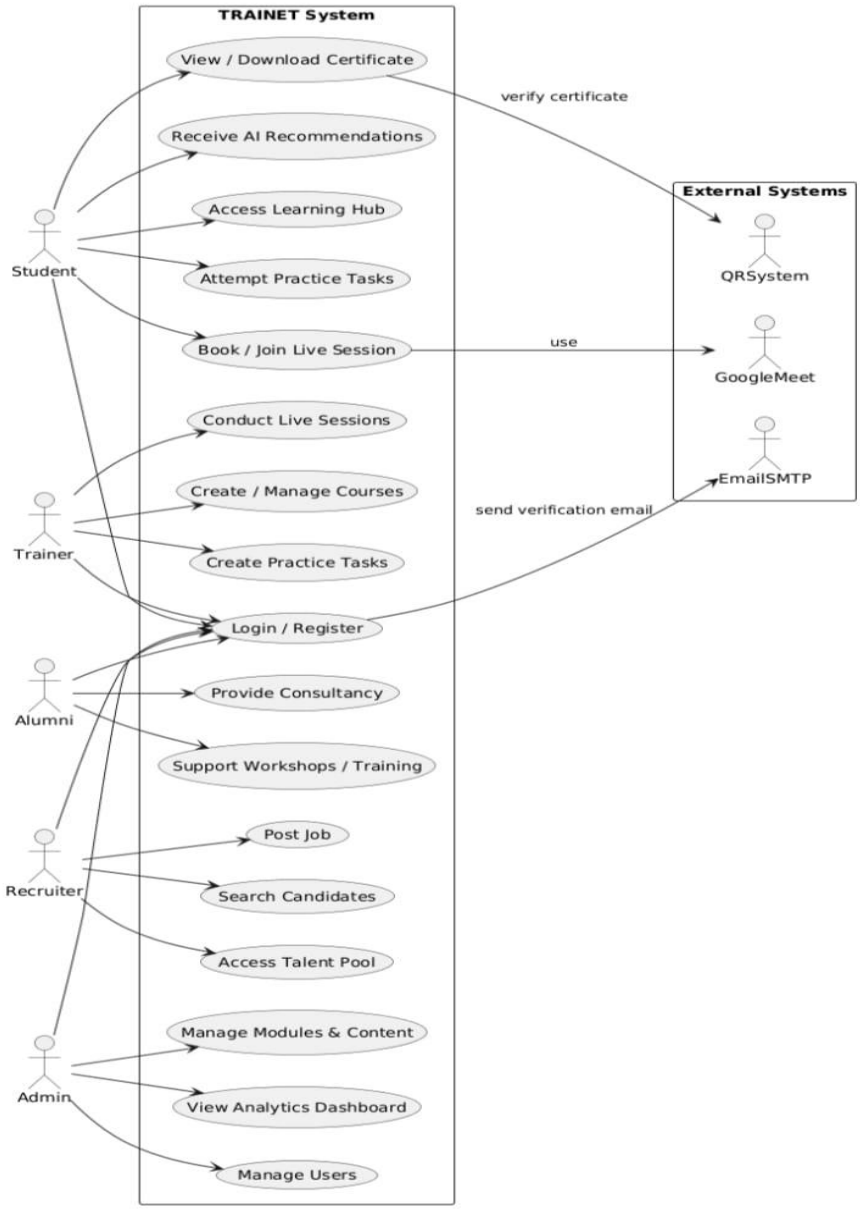

Figure 2: Use Case Diagram of TRAINET

Fully Dressed Use Cases

Item	Description
Use Case ID	UC-01
Use Case Name	Enroll in Course
Actors	Student
Preconditions	User must be logged in with a Student role.
Postconditions	Student is successfully enrolled in the selected course.
Trigger	Student chooses a course to join.
Main Flow	1. Student navigates to Course Catalog. 2. Student selects a course. 3. System displays enrollment methods. 4. Student enrolls using QR-based or direct enrollment. 5. System confirms enrollment and updates Student profile.
Alternate Flow	3a. Course is full → System displays "Course Full" message.
Exceptions	QR code is invalid → enrollment fails. Server unavailable → temporary error.

Table 9: USECASE of TRAINET

Item	Description
Use Case ID	UC-02
Use Case Name	Submit assignment
Actors	Student
Preconditions	The student must be enrolled in the course
Postconditions	Assignment is successfully submitted and ready for grading
Trigger	Student uploads an assignment file.
Main Flow	1. Student opens assignment section. 2. Student uploads file. 3. System performs plagiarism checks. 4. System stores submission. 5. System notifies Trainer.
Alternate Flow	2a. Student uploads unsupported file type → System rejects.
Exceptions	File upload fails due to poor internet.

Table 10: USECASE of TRAINET

Item	Description
Use Case ID	UC-03
Use Case Name	Generate Certificate
Actors	System/Student
Preconditions	Student has completed all course requirements.
Postconditions	A digital certificate with a QR code is generated.
Trigger	Course completion is marked.
Main Flow	1. System checks course completion status. 2. System generates certificate PDF. 3. System embeds unique QR code. 4. System stores certificate in student profile. 5. Student accesses downloadable certificate.
Exceptions	Service of certificate generation is offline

Table 11: USECASE of TRAINET

Item	Description
Use Case ID	UC-04
Use Case Name	Post Work & Practice Task
Actors	Trainer
Preconditions	Trainer is authenticated.
Postconditions	Task becomes visible to all enrolled students.
Trigger	Trainer creates a new practice task.
Main Flow	1. Trainer navigates to Work & Practice. 2. Trainer enters task details. 3. Trainer uploads instructions/resources. 4. System publishes the task.
Alternate flow	3a. Resource exceeds size limit → system prompts to reduce file size.

Table 12: USECASE of TRAINET

Item	Description
Use Case ID	UC-05
Use Case Name	Search candidates
Actors	Recruiters
Preconditions	Recruiter must be registered.
Postconditions	Recruiter receives a list of matched candidates.
Trigger	Recruiter enters skill requirements.

Main Flow	1. Recruiter opens Talent Pool. 2. Recruiter enters skills, experience, or project criteria. 3. System runs AI-based matching. 4. System displays matched learners. 5. Recruiter views learner profiles and portfolios.
Exceptions	No matching candidates found.

Table 13: USECASE of TRAINET

Item	Description
Use Case ID	UC-06
Use Case Name	Manage Users
Actors	Admin
Preconditions	Admin is authenticated.
Postconditions	User list is updated accordingly.
Trigger	Admin executes a management action.
Main Flow	1. Admin accesses dashboard. 2. Admin views user accounts. 3. Admin edits/deletes/approves roles. 4. System updates database.
Exceptions	Admin deletes a user linked to active courses → system blocks deletion.

Table 14: USECASE of TRAINET

3.3 External Interface Requirements

The interfaces that TRAINET uses are described in this section.

3.3.1 User Interfaces

With common browsers, TRAINET offers a contemporary, user-friendly, and role-specific online interface. The interface is intended to provide easy navigation, understandable material display, and uniform visual design throughout all sections.

1. **Interface for Students**
    - Dashboard displaying tasks, certifications, progress, and registered courses
    - Course sites include areas for uploading assignments, resources, and modules

- Task viewer and submission panel for work and practice
- Screen for certificate access and verification
- Page for profile settings

2. Interface for Trainers

- Interface for creating courses and uploading materials
- Creation of assignments and grading panel
- Posting page for work and practice tasks
- Screens that track students' progress

3. Interface for Alumni

- Experience, achievements, and skill areas in the profile builder
- Panel for arranging consultation and mentoring sessions
- Inbox for student communications

4. Interface for Recruiters

- Dashboard for Talent Pool
- Candidate search parameters (certifications, project kinds, and talents)
- Viewing screen for a portfolio
- Interface for messaging applicants

5. Interface for Administration

- System summary displayed on the admin dashboard
- Interface for user management
- Screens for module administration and configuration
- Viewer for security logs

3.3.2 Hardware Interfaces

- **Personal Computers:** Web browsers are used by desktop or laptop computers to access the platform.
- **Mobile devices:** Android or iOS-powered smartphones and tablets

Beyond common user devices, TRAINET does not require any additional hardware devices.

3.3.3 Software Interfaces

TRAINET's primary functions are supported by a number of software components. These interfaces specify how the system interacts with databases, operating systems, external software, and APIs.

Operating Systems interface

- TRAINET is a web-based system that operates on Windows, macOS, Linux, Android, iOS, and any other operating system that supports contemporary browsers.
- Other than regular browser compatibility, no OS-specific capabilities are needed.

Database Interface

- To store user accounts, course data, assignments, certificates, and system logs, TRAINET establishes a connection to a relational database (like MySQL or PostgreSQL).
- The database gets data in response to structured queries sent by the system.
- Among the shared data are:
    - Details about the user
    - Metadata for course materials
    - Submissions of assignments
    - Records of certificates
    - Tasks for Work and Practice

External Software / API Interfaces

The following software services are connected to by TRAINET:

1. Integration with Google Meet

- Used to create or include connections to live class sessions.
- Sending and receiving meeting URLs is a part of communication.

2. SMTP and email services

- Used to deliver alerts, OTPs, and verification emails.
- Among the outgoing messages are:
    - Emails for account verification
    - Messages to reset your password
    - System notifications

4. Compatibility with QR Scanners

- The system uses a web-enabled verification page in response to QR scan requests.

3. QR Code Generation Library

- Generates QR codes for certificates and course enrollments.
- Data passed to the library includes unique certificate IDs or enrollment IDs.

4. QR Scanner Compatibility

- System responds to QR scan requests via a web-enabled verification page.
- Incoming data: scanned QR code value.
- Outgoing data: certificate validity details.

Nature of Data Exchange

- When appropriate, RESTful requests are used in all communications.
- SQL is used for database operations, and JSON is utilized for API answers.
- There is no need for local program dependencies or global shared memory.

3.3.4 Communications Interfaces

The communication-related needs that TRAINET must support are outlined in this section. When the system communicates with users, external services, and internal components, it describes the data exchange standards, communication protocols, and security measures that are involved.

Communication Protocols

- For all client-server interactions, TRAINET will employ HTTP/HTTPS as the main communication channel.
- To guarantee secured transfer of sensitive data, including user information, certificates, and login passwords, HTTPS must be implemented.

Formatting Messages

- For REST API connectivity, data sent between the front-end and back-end must be in JSON format.
- To provide consistency and traceability, server answers or internal system logs must adhere to organized message formats.

Communication via Email and Notification

- The system will interact over SMTP in order to deliver verification emails, password reset links, OTP tokens, and general alerts,
- To maintain uniformity and clarity, all emails must adhere to standard formats.

Communication Requirements for Third Parties

- For live session connections, TRAINET will communicate with Google Meet using URL-based communication.
- When a QR code is scanned, a request is sent to the TRAINET verification endpoint via URL redirection.

Encryption and Security

- SSL/TLS encryption must be used for any communications that involve user data.
- Plain text should never be used to send sensitive information.
- To avoid unwanted tampering, certificate verification data must be encrypted.

Synchronization and Performance Requirements

There will be support for real-time communication for:

- Submissions of assignments
- Updates on course progress
- Access to a live session

To prevent data conflicts, the system must maintain constant synchronization throughout several user sessions.

4 Other Non-functional Requirements

4.1 Performance Requirements

The performance criteria listed below guarantee that TRAINET will continue to be responsive, scalable, and effective in both typical and high usage scenarios.

PR-1: Under typical network conditions, the system should load the dashboard and main pages in 3 to 5 seconds.
PR-2: No course-related transaction, such as accessing content or turning in assignments, may take more than 10 seconds.
PR-3: At least 100 concurrent users must be supported by the system without experiencing noticeable performance decrease.
PR-4: Depending on the size of the assignment, AI-based auto-grading processes must generate results in 60-120 seconds.
PR-5: The creation of a certificate with an integrated QR code must be finished in 30 seconds.
PR-6: The system must quickly manage real-time synchronization for task postings and assignment submissions.
PR-7: To guarantee constant accessibility, server uptime must be kept at 99% or greater.

4.2 Safety and Security Requirements

Data protection, safe system behaviour, and adherence to security standards are all covered by these criteria.

Safety Concerns
SR-1: Critical data (such as certificates and course records) cannot be deleted by the system without verification.
SR-2: Prior to irreversible actions like deleting a user or removing a task, the system must provide

warning messages.

SR-3: To avoid data loss, automated database record backups must be carried out on a regular basis.

SR-4: To handle unforeseen failures without causing system crashes, the system must have error-handling techniques.

Security Requirements

SEC-1: Secure hashing techniques, such as bcrypt and SHA-256, must be used to store all user passwords.

SEC-2: HTTPS and TLS encryption must be used to protect sensitive data while it is being sent.

SEC-3: To stop unwanted access to restricted modules, role-based access control must be implemented.

SEC-4: To verify newly registered accounts, the system will provide OTPs or verification links.

SEC-5: The system must guard against typical security risks such CSRF attacks, SQL injection, and cross-site scripting (XSS).

SEC-6: The system will only let authenticated trainers and graduates to publish information or hold sessions.

SEC-7: By verifying distinct certificate IDs on the server, QR-based certificate verification must guarantee authenticity.

SEC-8: For auditing reasons, security logs must document administrator operations, login attempts, and system modifications.

4.3 Software Quality Attributes

To guarantee usability, dependability, and long-term maintainability, TRAINET must adhere to fundamental software quality standards. Every characteristic listed below has a specific, verified requirement that is in line with your SRDS template.

4.3.1 Reliability

- The system must be able to withstand small network disruptions without corrupting data.
- For partial uploads or unsuccessful requests, automatic recovery procedures must be put in place.

4.3.2 Usability

- The navigation flow, button placement, and style of all user interfaces must be uniform.
- Without formal training, new users should be able to utilize the system with ease.

4.3.3 Maintainability

- Modular architecture will be used in the system's development so that any module may be upgraded or changed on its own.
- For upcoming improvements, code documentation and API references must be kept up to date.

4.3.4 Scalability

- Future development, such as the addition of additional modules, courses, or more users, must be supported by the system design.
- If necessary, the database structure must support horizontal scalability.

4.3.5 Portability

- Chrome, Edge, Firefox, and Safari are just a few of the contemporary browsers that TRAINET will work seamlessly with.
- Multiple device types (desktop, laptop, mobile, tablet) must be supported by the system without any interface breakdowns.

4.3.6 Interoperability

- The technology would enable smooth communication with other services including email servers, QR libraries, and Google Meet.
- To enable integration with upcoming third-party products, the system will make use of APIs.

4.3.7 Security (Quality-Level)

- Periodic vulnerability evaluations will be conducted on the system.
- All modules must have uniform enforcement of data access regulations.

4.3.8 Design for Change

- To guarantee that new features may be introduced with minimum impact on current functionalities, the system must be built with obvious module separation.

5 Design Description

5.1 Composite Viewpoint

Package diagram:

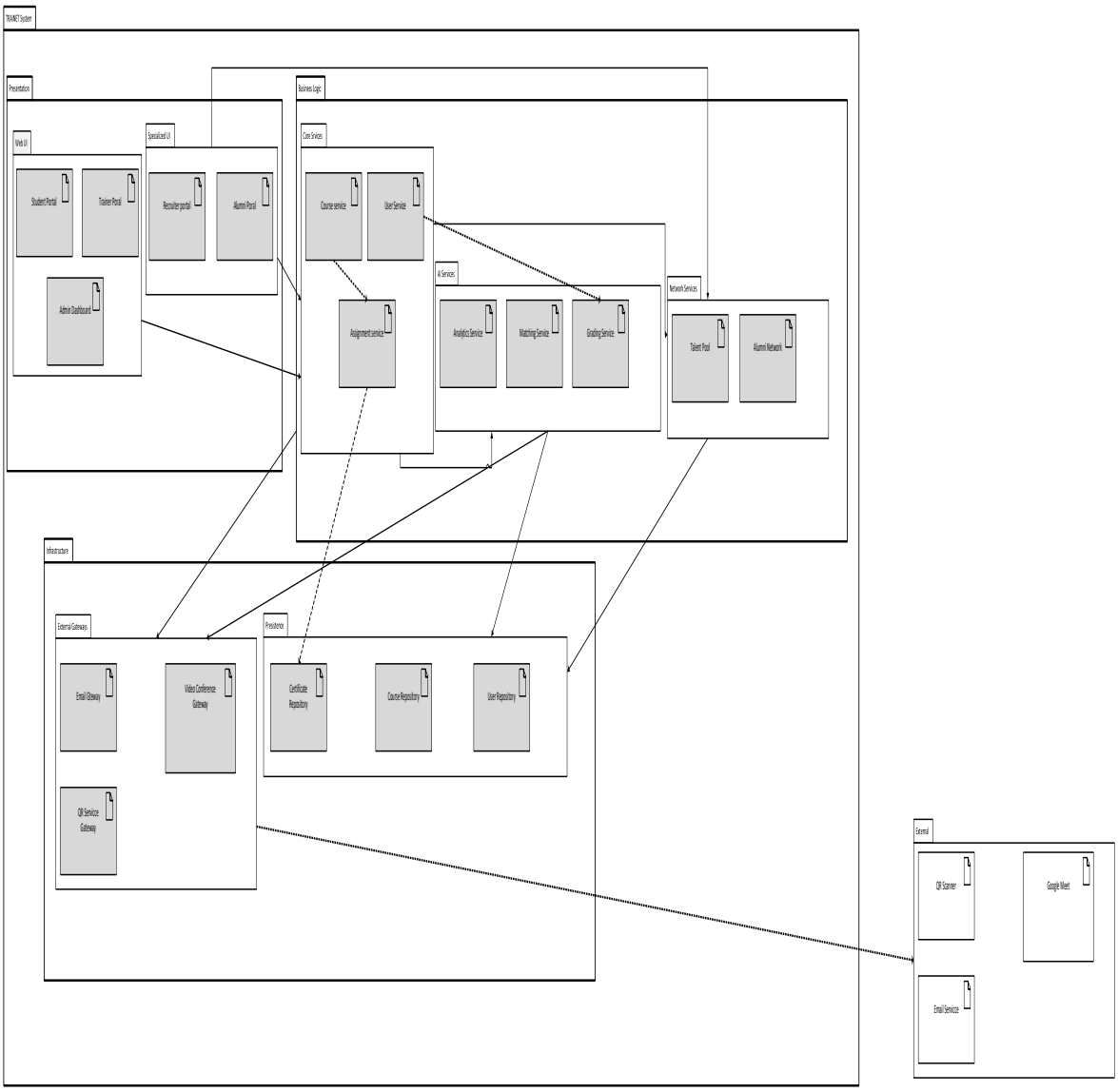

Figure 3: Package Diagram of TRAINET

5.2 Logical Viewpoint

Class Diagram:

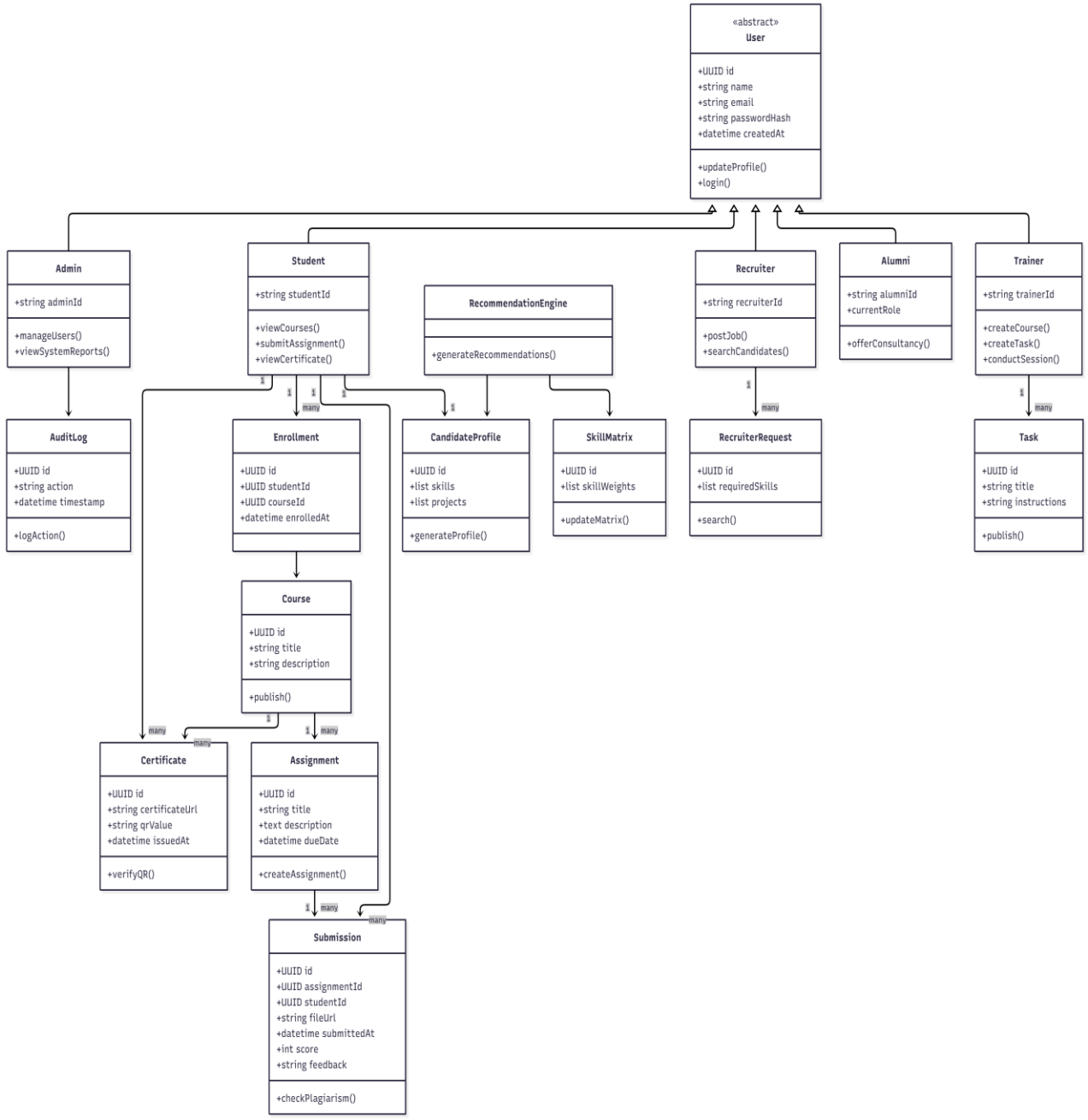

Figure 4: Class Diagram of TRAINET

5.3 Information Viewpoint

ERD (Entity Relationship Diagram):

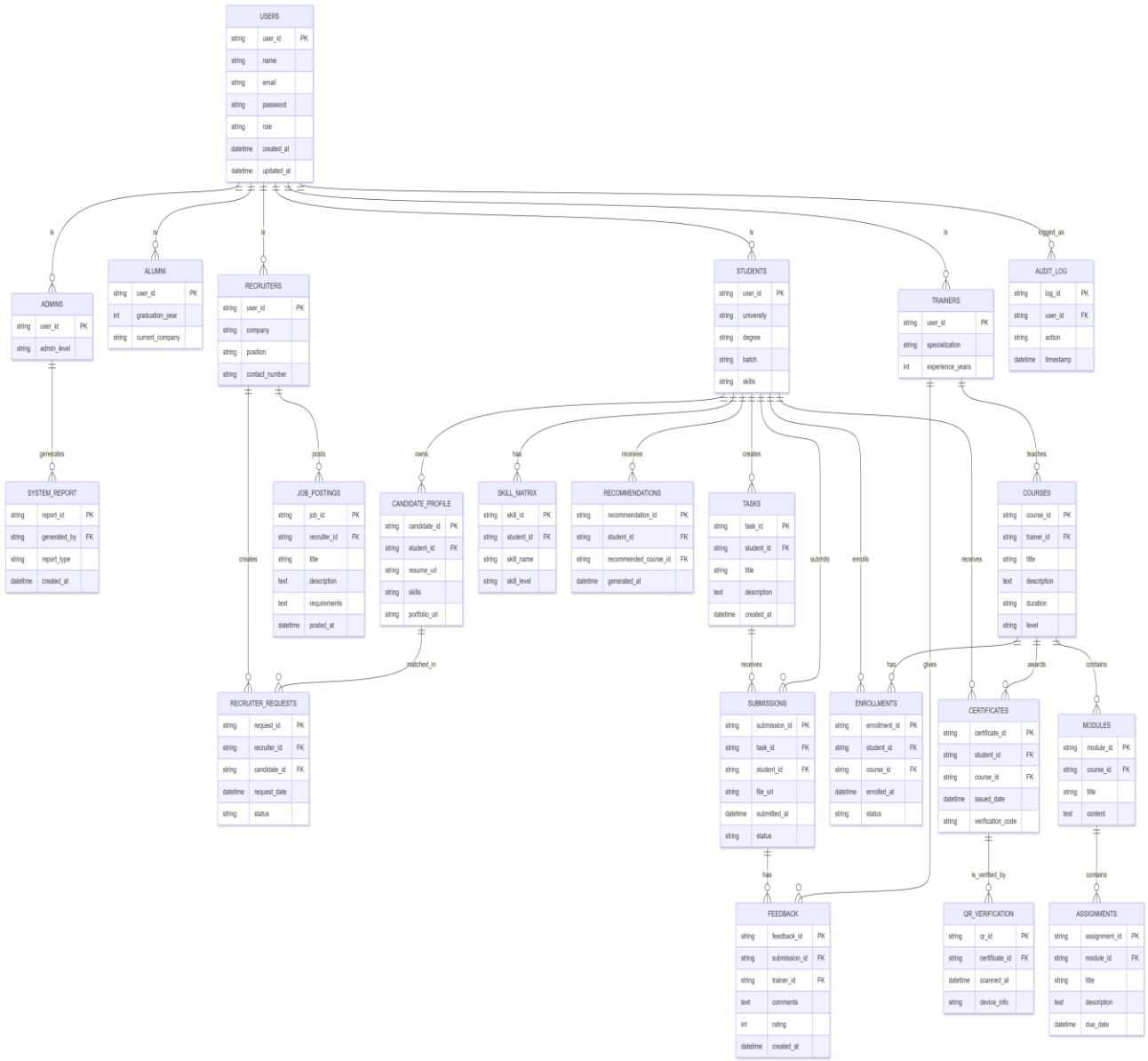

Figure 5: Entity Relationship Diagram of TRAINET

5.4 Interaction Viewpoint

5.4.1 Sequence: Enroll in Course

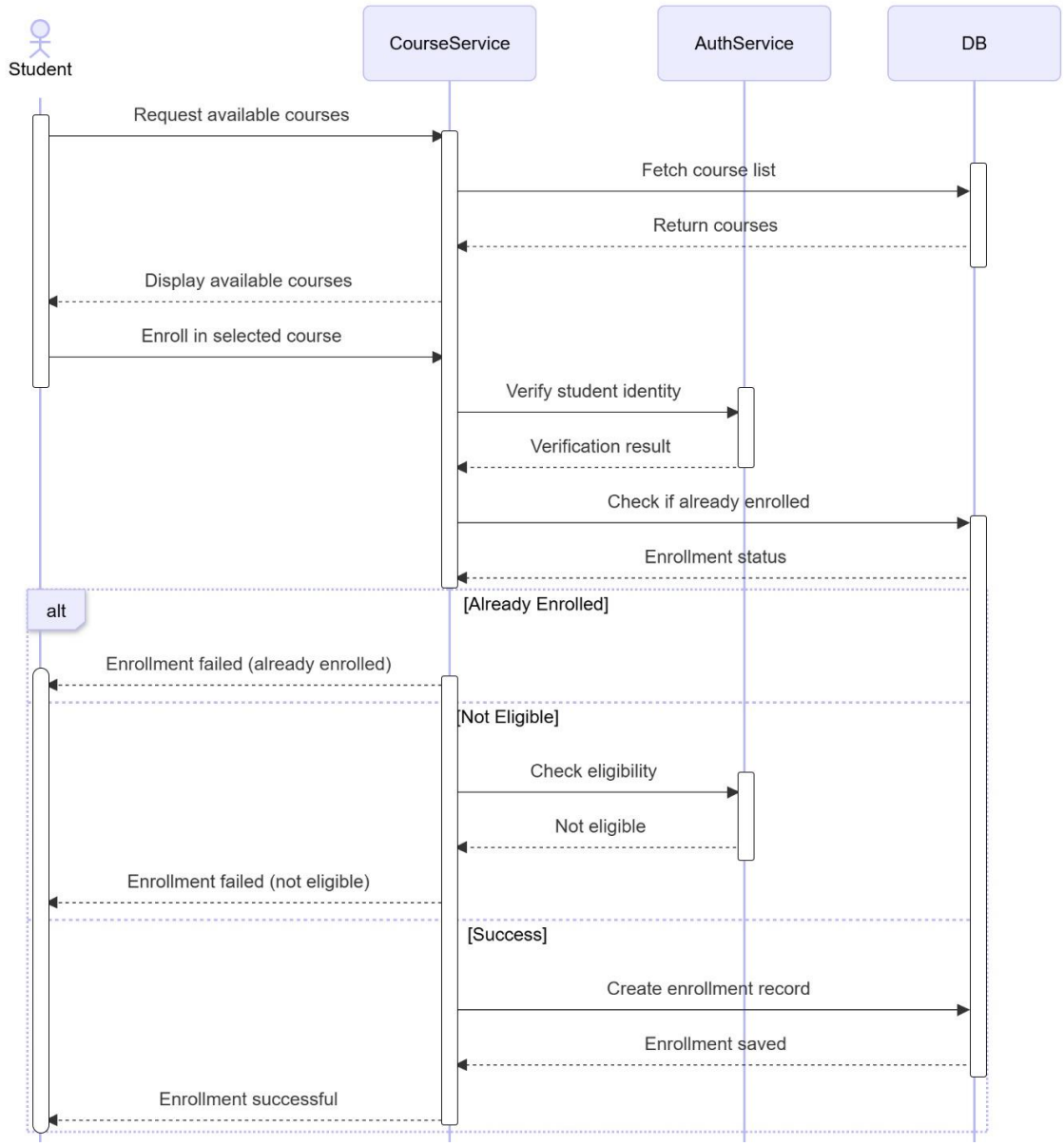

Figure 6: Sequence Diagram of TRAINET

5.4.2 Sequence: Submit Assignment (Plagiarism Check + Notification)

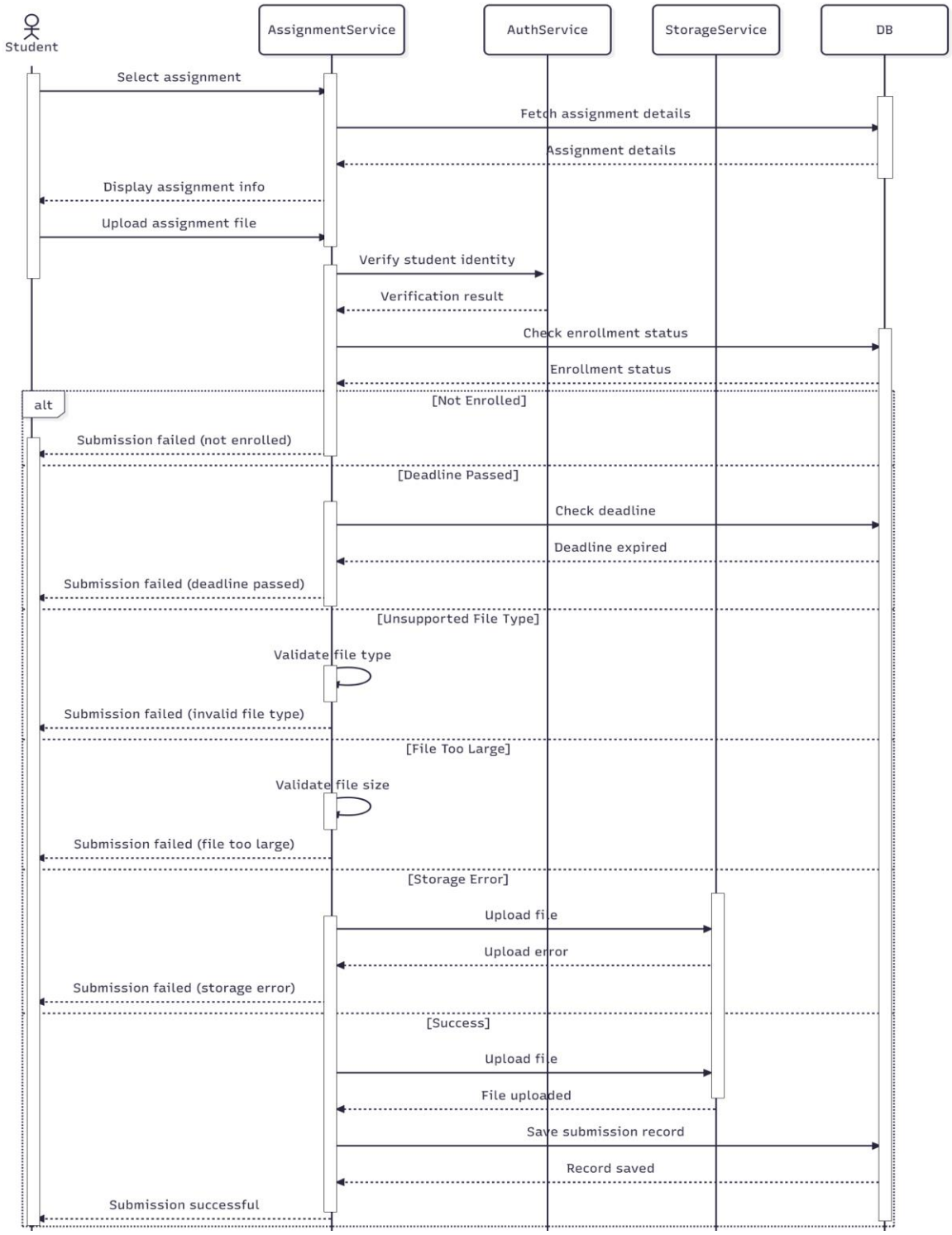

Figure 7: Sequence Diagram of TRAINET

5.4.3 Sequence: Generate Certificate (Course Completion)

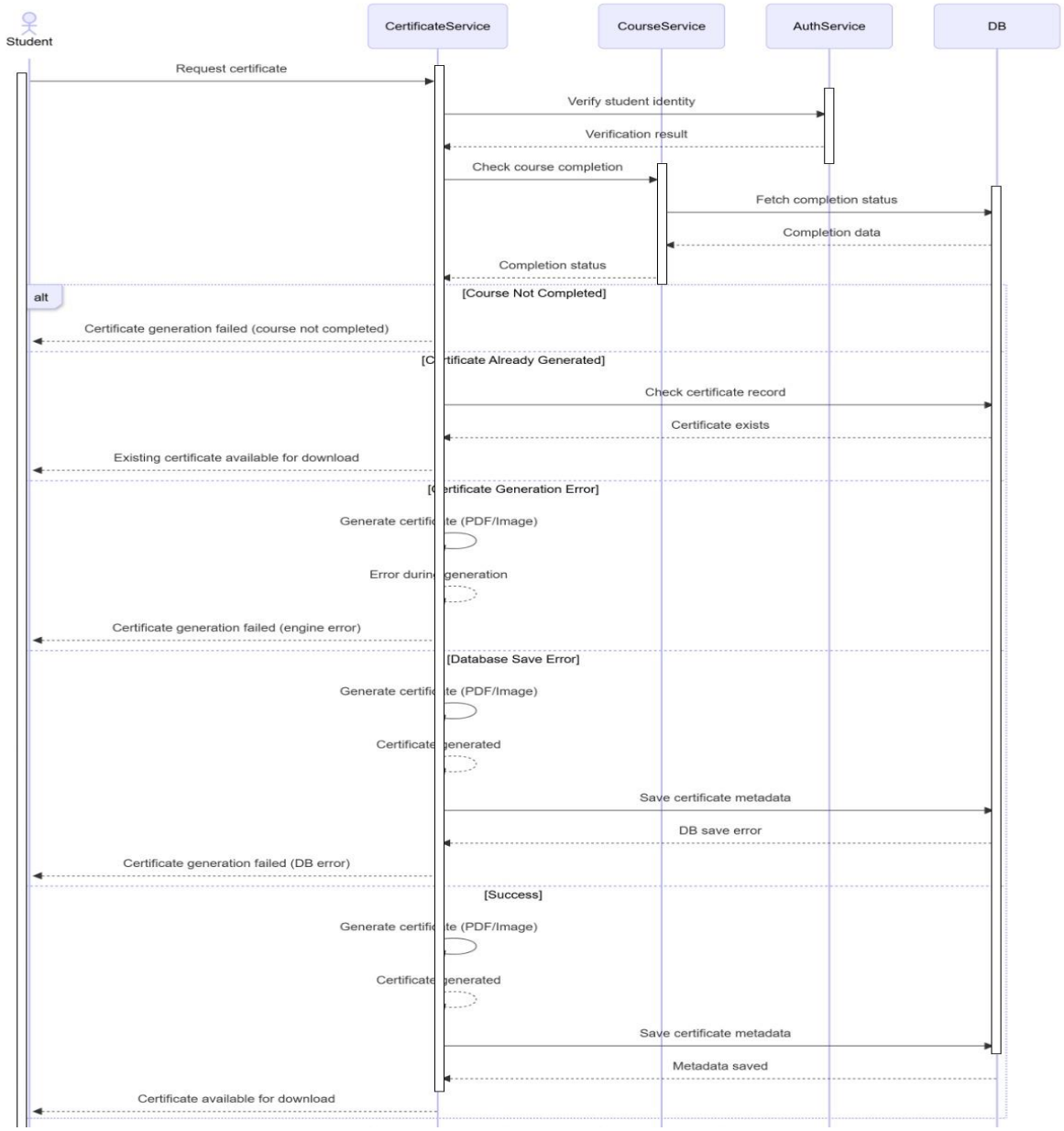

Figure 8: Sequence Diagram of TRAINET

5.4.4 Sequence: Post Work & Practice Task

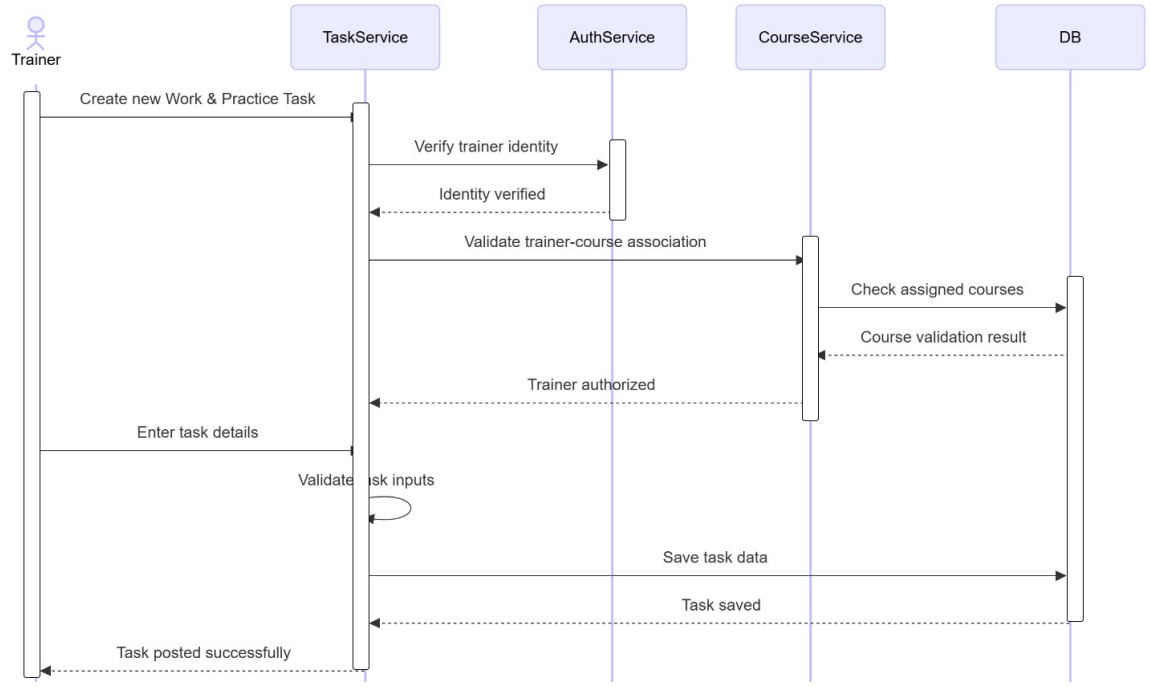

Figure 9: Sequence Diagram of TRAINET

5.4.5 Sequence: Search Candidates

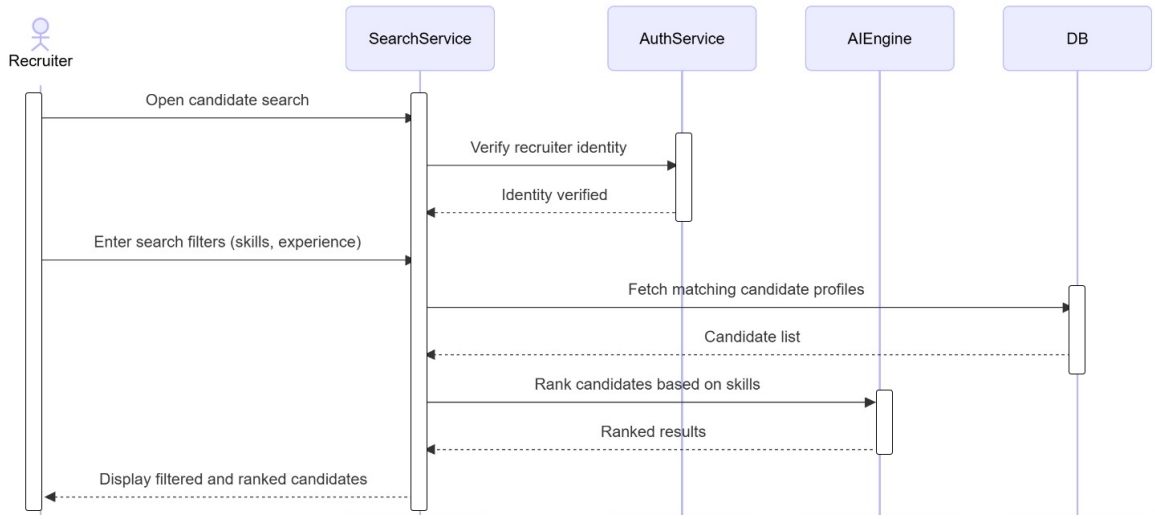

Figure 10: Sequence Diagram of TRAINET

5.4.6 Sequence: Manage Users (Admin)

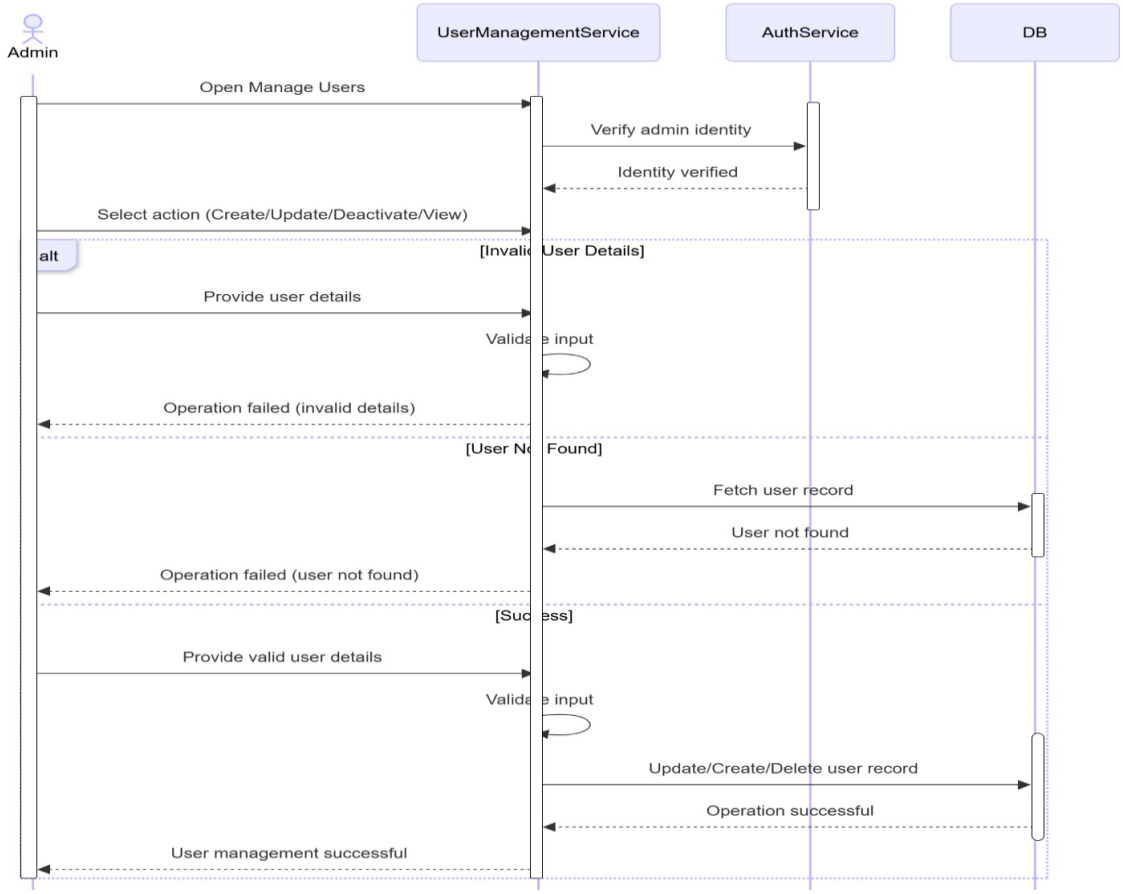

Figure 11: Sequence Diagram of TRAINET

5.5 State Dynamics Viewpoint

5.5.1 Assignment Submission

Figure 12: State Machine Diagram of TRAINET

5.5.2 Certificate

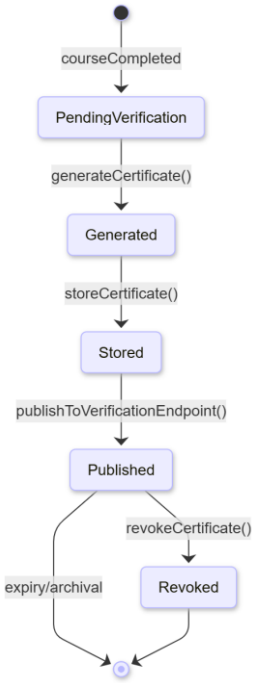

Figure 13: State Machine Diagram of TRAINET

5.5.3 Enrollment

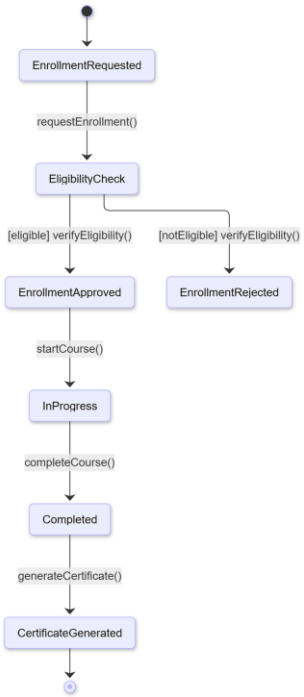

Figure 14: State Machine Diagram of TRAINET

5.5.4 Task (Work & Practice)

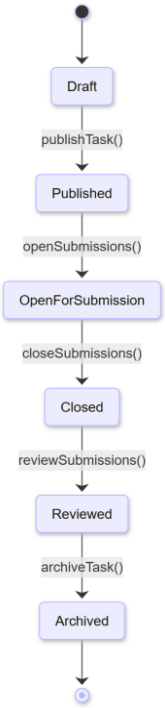

Figure 15: State Machine Diagram of TRAINET

5.6 Algorithm Viewpoint

The algorithmic design for important TRAINET system functions is described in this section. Pseudo code defines the logic, processes, and decision points for key functions, whereas decision tables are utilized to simplify complicated business rules with numerous circumstances.

5.6.1.1 Pseudo Code for Core Algorithms

Algorithm: AI-Powered Auto-Grading & Feedback

When a student submits an assignment, this algorithm is activated. For supported question kinds (such as coding problems and quizzes), it manages automatic grading and plagiarism detection.

Pseudo code:

```
FUNCTION autoGradeAssignment(submission_id, assignment_id)
    submission = GET submission FROM database WHERE id = submission_id
    assignment = GET assignment FROM database WHERE id = assignment_id
    plagiarism_report = CALL plagiarismDetectionService(submission.content)
    plagiarism_score = plagiarism_report.similarityPercentage
    IF plagiarism_score > assignment.allowedPlagiarismThreshold THEN
        SET submission.status = "Flagged for Plagiarism"
        SET submission.grade = 0
        SET submission.feedback = "Submission flagged for high similarity (" + plagiarism_score +
"%). Please review academic integrity policy."
        SAVE submission TO database
        RETURN // Exit early if plagiarism is too high
    END IF
    IF assignment.type == "Coding" THEN
        test_results = CALL codeEvaluationService(submission.content, assignment.test_cases)
        grade = CALCULATE grade BASED ON test_results.passedCases
        feedback = "Passed " + test_results.passedCases + " out of " + test_results.totalCases + " test
cases."
```

```
    ELSE IF assignment.type == "Quiz" THEN
        grade = COMPARE submission.answers WITH assignment.answer_key
        feedback = "Auto-graded based on answer key."

    ELSE
        grade = 0
        feedback = "Awaiting manual grading by trainer."
    END IF
    SET submission.grade = grade
    SET submission.feedback = feedback
    SET submission.status = "Graded"
    SET submission.graded_by = "AI System"
    SAVE submission TO database
    SEND NOTIFICATION to assignment.trainer: "Submission #" + submission_id + " has been
auto-graded."
END FUNCTION
```

Algorithm: AI-Based Candidate Matching

When a recruiter looks for applicants in the Talent Pool, this algorithm is used. It compares students' project experiences and skill sets to the recruiter's requirements.

Pseudo code:

```
FUNCTION findCandidates(recruiter_id, search_criteria)
    required_skills = search_criteria.skills
    min_experience = search_criteria.minExperience
    project_type = search_criteria.preferredProjectType
    potential_candidates = GET learners FROM database WHERE profile.is_in_talent_pool = True
    matched_candidates = []
    FOR EACH candidate IN potential_candidates DO
        score = 0
        candidate_skills = candidate.profile.skills
        matched_skill_count = COUNT OF required_skills THAT ARE IN candidate_skills
        skill_score = (matched_skill_count / LENGTH OF required_skills) * 50
        score = score + skill_score
```

```
        relevant_projects = GET projects FROM candidate WHERE type = project_type
        IF relevant_projects EXISTS THEN
            project_score = 30
        ELSE
            project_score = 0
        END IF
        score = score + project_score
        performance_score = (candidate.profile.average_grade / 100) * 20 // Assuming grade is out of
100
        score = score + performance_score
        IF candidate.profile.experience_level >= min_experience THEN
            ADD {candidate: candidate, match_score: score} TO matched_candidates
        END IF
    END FOR
    SORT matched_candidates BY match_score DESCENDING
    RETURN matched_candidates
END FUNCTION
```

Algorithm: QR Certificate Verification

When a QR code on a certificate is scanned by an external user, this algorithm is run. It confirms the legitimacy of the certificate.

Pseudo code

```
FUNCTION verifyCertificate(qr_code_data)
    certificate_uuid = DECODE qr_code_data
    certificate = GET certificate FROM database WHERE uuid = certificate_uuid
    IF certificate IS NULL THEN
        RETURN { status: "INVALID", message: "Certificate not found in the system." }
    END IF
        RETURN { status: "REVOKED", message: "This certificate is no longer valid." }
    END IF
    learner_name = certificate.learner.name
    course_name = certificate.course.name
    issue_date = certificate.issue_date

    RETURN {
        status: "VALID",
        message: "This is a verified certificate issued by TRAINET.",
```

```
        details: {
            learner: learner_name,
            course: course_name,
            issued_on: issue_date
        }
    }

END FUNCTION
```